Stackelberg Game-Based Vehicle-Aided Task Offloading for UAVs

Jihyun Park, Changkyung Kim, and Sukyoung Lee

***Abstract*—Owing to the increasing number of requests from smart devices to mobile edge computing servers, unmanned aerial vehicles (UAVs) have emerged as alternatives for providing computing resources to these devices. However, satisfying the delay requirements for all smart-device requests is challenging because UAVs have limited computing resources. To address this problem, we propose a vehicle-aided task offloading scheme for UAVs. We designed a Stackelberg game model to determine the optimal UAV demand and vehicle resource prices. Simulation results show that the proposed scheme completes the requested tasks faster than other benchmark methods.**

***Index Terms*—Delay, Stackelberg game, task offloading, unmanned aerial vehicles (UAVs), utility, vehicle.**

I. Introduction

THE RISE of smart applications has made fulfilling delay requirements challenging due to the constrained computing resources of Internet of Things (IoT) devices. Mobile edge computing (MEC) allows IoT devices to offload tasks to MEC servers, reducing task processing delays. Nevertheless, the significant number of service requests from IoT devices in smart cities can overwhelm MEC servers [1]. Additionally, their computing and communication resources may be compromised during natural disasters [2]. In such cases, unmanned aerial vehicles (UAVs) can process these tasks, but their limited computing resources make it difficult to meet all delay requirements. Hence, some studies have proposed offloading UAVs' computations to vehicles to improve the processing of these tasks, satisfying their delay requirements [3], [4]. However, private vehicles are unwilling to process such tasks to save computational resources and energy [5]. Several previous studies have proposed incentive mechanisms to determine an optimal task offloading strategy while incentivizing vehicles to provide their computing resources using game theory [3], [6]. However, these studies did not consider optimizing the price of computing resources based on the market demand.

This letter aims to determine an optimal task offloading strategy from the market perspective for UAVs requesting resources from vehicles by modeling their interactions as a two-stage Stackelberg game. Then, we prove that the game has a unique Nash equilibrium and offers optimal task offloading for UAVs. Finally, the simulation results reveal that the proposed algorithm outperforms other benchmark schemes in terms of the delay in completing the requested tasks.

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II. System Model

The system architecture of the proposed scheme comprises a set of UAVs $u \in U$ and vehicles $g \in G$. A system is considered in which UAVs are dispatched to an area to support MEC in processing offloaded tasks from nearby IoT devices [3], [7]. Each UAV *u* aims to maximize its computing resource purchases from vehicles within its budget, whereas each vehicle *g* aims to maximize its revenue by offering an idle computing resource. The payment for resources can be made in various forms, such as service fees, credits, or digital tokens [8]. We assume that this payment method has been previously agreed upon between UAVs and vehicles.

The UAV communicates with vehicles via an air-to-ground (A2G) downlink to offload tasks. The position coordinates of UAV u and vehicle g are denoted by (x_u, y_u, h_u) and (x_g, y_g, h_g), respectively, where (x_u, y_u) and (x_g, y_g) represent the horizontal coordinates, h_u indicates the flying altitude of UAV *u*, and h_g is the height of vehicle *g*. The distance between UAV *u* and vehicle *g* is calculated as $d_{u,g} = \sqrt{(x_u - x_g)^2 + (y_u - y_g)^2 + (h_u - h_g)^2}$. The UAV establishes line-of-sight (LoS) links with vehicles and experiences small-scale fading due to the scattering components. The channel between the UAV and each vehicle can be modeled as $H_g = \sqrt{\sigma d_{u,g}^{-2}} H_r$, where $\sigma d_{u,g}^{-2}$ indicates the large-scale channel power gain, σ denotes the channel power gain at a reference distance of $d_0 = 1$ m, and H_r represents the small-scale fading coefficient. The Rician channel is adopted to model small-scale fading as $H_r = \sqrt{K/(K+1)} H_l + \sqrt{1/(K+1)} H_n$, where H_l denotes LoS component satisfying $|H_l| = 1$, H_n denotes the random scattered component obeying the circularly symmetric complex Gaussian (CSCG) distribution with a zero-mean and unit-variance, and *K* is the Rician factor [9]. The UAV *u* sends data to vehicle *g* with power $P_{u,g}$, and the received power at vehicle *g* is $|H_r|^2 \sigma d_{u,g}^{-2} P_{u,g}$. As each UAV is allocated with the orthogonal channel [10], the signal-to-noise ratio (SNR) from UAV *u* to vehicle *g* is computed as $\gamma_{u,g} = P_{u,g} |H_r|^2 \sigma d_{u,g}^{-2} / B N_0$, where *B* represents the channel bandwidth of the A2G link, and N_0 indicates the noise power spectral density. Thus, the available transmission rate between UAV *u* and vehicle *g* is $R_{u,g} = B \log_2(1 + \gamma_{u,g})$.

A task generated by UAV *u* is expressed as (s_u, ρ_u), where s_u and ρ_u are the data size and processing capacity required for the task, respectively. The demand of UAV *u* for computing resources from each vehicle *g* is denoted by $x_{u,g}$, which is determined using the proposed algorithm (Section III) and indicates the quantity of resources to be purchased from vehicle *g*. Once the value of $x_{u,g}$ is determined, the transmission time from UAV u to vehicle g is computed as $t_{u,g}^{tx} = \frac{s_{u,g}}{R_{u,g}}$ and the time required by vehicle g to process the task as $t_{u,g}^{c} = \frac{\rho_{u,g}}{x_{u,g}}$, where $s_{u,g}$ and $\rho_{u,g}$ represent the corresponding notation for s_u and ρ_u for vehicle g. Then, we obtain $s_{u,g} = \frac{s_u x_{u,g}}{\sum_g x_{u,g}}$ and $\rho_{u,g} = \frac{\rho_u x_{u,g}}{\sum_g x_{u,g}}$. The offloading delay for UAV u to complete the task is computed as follows:

$$t_u^o = \max_{\forall g \in G} \left(t_{u,g}^{tx} + t_{u,g}^{c} \right). \tag{1}$$

III. Game Formulation

With the incentives in the offloading process, involving adjusting prices based on market demand and supply, the offloading process between UAVs and vehicles can be modeled as a Stackelberg game, where a UAV is a buyer seeking to purchase the required computing resources from a vehicle that sells them. The utility function U_u of UAV u is defined as

$$U_u = \sum_{g \in G} R_{u,g} log\left(1 + x_{u,g}\right), \quad \forall u \in U, \tag{2}$$

where the log function is applied because it captures the diminishing marginal returns of the utility function [11]. The utility function for vehicle g is defined as

$$U_g\left(y_g, Y_{-g}\right) = y_g \sum_{u \in U} x_{u,g}, \tag{3}$$

where y_g denotes the unit price set by vehicle g, and Y_{-g} represents the price matrix for all other vehicles except g.

A. UAV-Side Analysis

Each UAV aims to maximize its utility by purchasing the maximum computing resources, leading to a reduction in the offloading delay t_u^o, determined by $x_{u,g}$. The UAV's utility optimization problem can then be formulated as follows:

$$\max_{\{x_{u,g},\ \forall u \in U\}} \quad U_u, \tag{4}$$

$$s.t. \quad \sum_{g \in G} y_g x_{u,g} \leq C_u, \tag{5}$$

$$x_{u,g} \geq 0, \forall g \in G. \tag{6}$$

Eq. (5) ensures that the computing resources purchased by UAV u from all vehicles do not exceed C_u, the budget for purchasing computing resources from vehicles. Because this utility optimization problem is convex, the stationary solution is unique and optimal. First, we derive a reaction function for $|U|$ UAVs and two vehicles. In this case, the optimization problem for the UAV is expressed as

$$\max_{\{x_{u,1}, x_{u,2}\}} \quad U_u = \sum_{g=1,2} R_{u,g} log\left(1 + x_{u,g}\right), \tag{7}$$

$$s.t. \quad \sum_{g=1,2} y_g x_{u,g} \leq C_u, \tag{8}$$

$$x_{u,1}, x_{u,2} \geq 0. \tag{9}$$

For a convex optimization problem, the Karush–Kuhn–Tucker (KKT) conditions constitute necessary and sufficient conditions for the optimal solution [11]. Eqs. (7)–(9) can always have a feasible solution by setting the values of $x_{u,1}$ and $x_{u,2}$ as positive and sufficiently small such that Eqs. (8)–(9) are satisfied. This indicates that a strong duality exists, and Slater's condition holds. We define $\mu_{u,1}$, $\mu_{u,2}$, and $\mu_{u,3}$ as the Lagrange's multipliers associated with Eqs. (8)–(9). The Lagrangian function is expressed as

$$L_u = \sum_{g=1,2} R_{u,g} log\left(1 + x_{u,g}\right) + \mu_{u,1} \left(C_u - \sum_{g=1,2} y_g x_{u,g} \right) + \mu_{u,2} x_{u,1} + \mu_{u,3} x_{u,2}. \tag{10}$$

The following are the KKT conditions for Eqs. (7)–(9):

$$\partial L_u / \partial x_{u,1} = R_{u,1} / \left(1 + x_{u,1}\right) - \mu_{u,1} y_1 + \mu_{u,2} = 0, \tag{11}$$

$$\partial L_u / \partial x_{u,2} = R_{u,2} / \left(1 + x_{u,2}\right) - \mu_{u,1} y_2 + \mu_{u,3} = 0, \tag{12}$$

$$\mu_{u,1} \left(C_u - \sum_{g=1,2} y_g x_{u,g} \right) = 0, \tag{13}$$

$$\mu_{u,2} x_{u,1} = 0, \tag{14}$$

$$\mu_{u,3} x_{u,2} = 0, \tag{15}$$

$$\mu_{u,1} > 0,\ \mu_{u,2}, \mu_{u,3}, x_{u,1}, x_{u,2} \geq 0. \tag{16}$$

The optimal demand for UAVs can be expressed as one of the following cases:

1) Case 1: $x_{u,1} = x_{u,2} = 0$: When $C_u = 0$, the UAV cannot afford computing resources, $x_{u,1} = x_{u,2} = 0$ is clear. If $C_u > 0$, $\mu_{u,1} = 0$ using Eq. (13), and $\mu_{u,2}$ and $\mu_{u,3}$ are negative according to Eqs. (11)–(12), respectively, which conflict with Eq. (16). Therefore, this situation cannot occur.

2) Case 2: $x_{u,1}, x_{u,2} > 0$: In this case, using Eqs. (14)–(15), we derive $\mu_{u,2} = \mu_{u,3} = 0$. Then, Eqs. (11)–(13) can be re-expressed as follows:

$$R_{u,1} / \left(1 + x_{u,1}\right) - \mu_{u,1} y_1 = 0, \tag{17a}$$

$$R_{u,2} / \left(1 + x_{u,2}\right) - \mu_{u,1} y_2 = 0, \tag{17b}$$

$$\mu_{u,1} \left(C_u - \sum_{g=1,2} y_g x_{u,g} \right) = 0. \tag{17c}$$

From Eqs. (17a)–(17c), we derive

$$x_{u,g} = \frac{R_{u,g}}{\mu_{u,1} y_g} - 1, \forall u \in U, \quad g = 1, 2, \tag{18}$$

$$\frac{1}{\mu_{u,1}} = \frac{C_u + \sum_{g=1}^{2} y_g}{\sum_{g=1}^{2} R_{u,g}}. \tag{19}$$

Then, substituting Eq. (19) into Eq. (18) yields:

$$x_{u,g} = \frac{R_{u,g} \left(C_u + \sum_{g=1}^{2} y_g \right)}{y_g \sum_{g=1}^{2} R_{u,g}} - 1, \quad g = 1, 2. \tag{20}$$

3) Case 3: $x_{u,1} > 0$, $x_{u,2} = 0$: Eq. (14) implies that $\mu_{u,2} =$ 0, and from Eq. (11), we derive

$$x_{u,1} = R_{u,1} / \left(\mu_{u,1} y_1 \right) - 1. \tag{21}$$

By substituting $x_{u,1}$ into Eq. (13),

$$\mu_{u,1} \left(C_u - \frac{R_{u,1}}{\mu_{u,1}} + y_1 \right) = 0, \tag{22}$$

and because $\mu_{u,1} > 0$, we can derive $\mu_{u,1} = \frac{R_{u,1}}{C_u + y_1}$ from Eq. (22). By substituting $\mu_{u,1}$ into Eq. (21), we obtain

$$x_{u,1} = \left(C_u + y_1 \right) / y_1 - 1 = C_u / y_1. \tag{23}$$

Because $x_{u,2} = 0$, we obtain $\frac{R_{u,2}(C_u + \sum_{g=1}^{2} y_g)}{y_2 \sum_{g=1}^{2} R_{u,g}} = 1$, $C_u = (y_2 R_{u,1} - y_1 R_{u,2}) / R_{u,2}$ from Eq. (20). Using this, we can

Algorithm 1 Optimal UAV Demand Calculation

1: **Input:** C_u, R_{min}, $S = G$, S_N=$\{g|y_g > 0, x_{u,g} = 0\} = \emptyset$;
2: **Output:** optimal $x_{u,g}$;
3: **for** $u \in U$ **do**
4: g=1;
5: **while** $g \leq |G|$ **do**
6: **if** ($y_g = 0$ **or** $g \in S_N$) **then** $x_{u,g} = 0$;
7: **else**
8: $S' = \{g \in (S - S_N) \mid y_g \neq 0\}$;
9: $x_{u,g} = \frac{R_{u,g}(C_u + \sum_{g\in S'} y_g)}{y_g \sum_{g\in S'} R_{u,g}} - 1$;
10: **if** $x_{u,g} < 0$ **or** $R_{u,g} < R_{min}$
11: **then** $x_{u,g} = 0$, $S_N = S_N \cup \{g\}$;
12: g=g+1;
13: **return** $x_{u,g}$;

Algorithm 2 Finding the Nash Equilibrium

Input: threshold ϵ, δ, **Output:** optimal x^*, y^*;
Each vehicle g initializes price y_g;
while at least one vehicle changes the price **do**
Each UAV u calculates $x_{u,g}$ using Algorithm 1;
for $g \in G$ **do**
if $\sum_{u\in U} x_{u,g} < f_g - \epsilon$ **then**
Decrease the price $y_g' = y_g - \delta$;
Update the demand from all UAVs;
else if $\sum_{u\in U} x_{u,g} > f_g + \epsilon$ **then**
Increase the price $y_g' = y_g + \delta$;
Update the demand from all UAVs;
else price y_g unchanged;
if $|y_g' - y_g| < \epsilon$ **then**
$x^* = x_{u,g}$, $y^* = y_g'$
break;
return x^*, y^*

rewrite Eq. (23) as follows:

$$x_{u,1} = \frac{C_u}{y_1} = \frac{C_u\big(R_{u,1} + R_{u,2}\big)}{y_1\big(R_{u,1} + R_{u,2}\big)} = \frac{C_u R_{u,1} + R_{u,2}\big(\frac{y_2 R_{u,1} - y_1 R_{u,2}}{R_{u,2}}\big)}{y_1\big(R_{u,1} + R_{u,2}\big)} = \frac{R_{u,1}\Big(C_u + \sum_{g=1}^{2} y_g\Big)}{y_1 \sum_{g=1}^{2} R_{u,g}} - 1. \tag{24}$$

4) Case 4: $x_{u,1} = 0$, $x_{u,2} > 0$: Similar to Case 3,

$$x_{u,2} = \frac{C_u}{y_2} = \frac{R_{u,2}\Big(C_u + \sum_{g=1}^{2} y_g\Big)}{y_2 \sum_{g=1}^{2} R_{u,g}} - 1. \tag{25}$$

From Eqs. (20), (24), and (25), the optimal demand for the general case of $|U|$ UAVs and $|G|$ vehicles for a price set $\{y_g\}$ can be formulated as

$$x_{u,g} = \frac{R_{u,g}\Big(C_u + \sum_{g\in G} y_g\Big)}{y_g \sum_{g\in G} R_{u,g}} - 1, \tag{26}$$

where $x_{u,g} \geq 0, \forall g \in G, u \in U$. In addition, UAV u demands $x_{u,g} \geq 0$ from vehicle g if

$$y_g \leq \frac{R_{u,g}\Big(C_u + \sum_{k\in G, k\neq g} y_k\Big)}{\sum_{g\in G} R_{u,g} - R_{u,g}}. \tag{27}$$

The proposed algorithm to determine the optimal demand for $x_{u,g}$ is presented in Algorithm 1. UAV demand cannot be negative, $x_{u,g}$ is reset to zero when $x_{u,g}$ computed using Eq. (26) is negative (Lines 10–11). In addition, the UAV purchases $x_{u,g}$ only when $R_{u,g} \geq R_{\min}$, where $R_{\min}$ is the minimum required transmission rate for offloading to prevent a prolonged offloading delay. Algorithm 1 includes two loop levels, where the number of outer loop executions is $|U|$ and the number of inner loop executions is $|G|$, resulting in a time complexity of $O(|U||G|)$.

B. Vehicle-Side Analysis

Given Y_{-g}, each vehicle g selects an appropriate price y_g to maximize its utility. This utility maximization problem for vehicle g is formulated as follows:

$$\max_{\{y_g,\ \forall g\in G\}} \ U_g\big(y_g, Y_{-g}\big) = y_g \sum_{u\in U} x_{u,g}, \tag{28}$$

$$s.t. \ \sum_{u\in U} x_{u,g} \leq f_g, \tag{29}$$

$$y_g \geq 0, \forall g \in G, \tag{30}$$

where f_g denotes the idle computing resources of vehicle g, which refers to the remaining resources after the vehicle has used them for its own purposes. Substituting Eq. (26) into Eqs. (28)–(29) yields the following:

$$\max_{\{y_g,\ \forall g\in G\}} \ U_g\big(y_g, Y_{-g}\big) = y_g \sum_{u\in U} \Big(\frac{R_{u,g}\Big(C_u + \sum_{g\in G} y_g\Big)}{y_g \sum_{g\in G} R_{u,g}} - 1\Big), \tag{31}$$

$$s.t. \ \sum_{u\in U} \Big(\frac{R_{u,g}\Big(C_u + \sum_{g\in G} y_g\Big)}{y_g \sum_{g\in G} R_{u,g}} - 1\Big) \leq f_g. \tag{32}$$

The objective function $U_g(y_g, Y_{-g})$ indicates that the optimal price y_g for vehicle g is related to the prices of the other vehicles. Vehicles select price strategies to maximize utility, and this price optimization game is denoted by $\Gamma = (G, Y_G, U_G)$, where Y_G is the set of all possible price strategy matrices that satisfy the constraints of Eqs. (30) and (32), and U_G is the set of utilities associated with the vehicles. Given Y_{-g}, the idle computing resources that vehicle g can sell are limited by Eq. (32). Thus, we can obtain

$$y_g \geq \sum_{u\in U} \frac{R_{u,g}\Big(C_u + \sum_{k\in G, k\neq g} y_k\Big)}{\sum_{g\in G} R_{u,g}\big(f_g + 1\big) \ - R_{u,g}}. \tag{33}$$

Definition 1 (Nash Equilibrium): The price strategy matrix $y_g^* \in Y_G$ is the Nash equilibrium of game Γ if no vehicle can increase its utility further by unilaterally changing its strategy.

$$U_g\big(y_g^*, y_{-g}^*\big) \geq U_g\big(y_g, y_{-g}^*\big), \forall g \in G.$$

The Nash equilibrium has self-stabilizing properties that allow for mutual satisfaction; therefore, it does not offer any incentive for the vehicles in the equilibrium to change.

Theorem 1: A unique Nash equilibrium of the price optimization game always exists among vehicles.

Proof: In game Γ, the strategy sets of all vehicles $Y = Y_1 \times Y_2 \times \cdots \times Y_G$ are non-empty, convex, and closed subsets of some Euclidean space $\mathbb{R}^G$. The utility function U_g is continuous in y_g; hence, Eq. (31) with respect to y_g can be rewritten as follows:

$$U_g(y_g, Y_{-g}) = \sum_{u \in U} \left(\frac{R_{u,g}(C_u + \sum_{g' \neq g} y_{g'})}{\sum_{g \in G} R_{u,g}} + \frac{R_{u,g}}{\sum_{g \in G} R_{u,g}} y_g - y_g \right). \quad (34)$$

The first-order derivative of U_g with respect to y_g is

$$\frac{\delta U_g\left(y_g, Y_{-g}\right)}{\delta y_g} = \sum_{u \in U} \left(\frac{R_{u,g}}{\sum_{g \in G} R_{u,g}} - 1 \right) < 0, \quad (35)$$

and its second-order derivative with respect to y_g is

$$\frac{\delta^2 U_g}{\delta y_g^2} = 0, \forall g \in G. \quad (36)$$

Eqs. (35)–(36) imply that $U_g(y_g, Y_{-g})$ is a linear function in y_g with a negative gradient. Therefore, vehicle g chooses the least possible value of y_g to maximize its utility. Given y_{-g}^*, the optimal price strategies of the other vehicles, y_g is bounded by Eqs. (27) and (33). The optimal price of the vehicle, y_g^*, is uniquely determined to be the least value that satisfies Eq. (33); thus, the price optimization game converges to a Nash equilibrium, where all vehicles select the respective minimal y_g^*. The uniqueness of y_g^* for each vehicle leads to a consistent and unique Nash equilibrium for the game. ∎

Definition 2 (Stackelberg Equilibrium): Set (Y^*, X^*) is referred to as the Stackelberg equilibrium of game Γ between the vehicles and UAVs if it satisfies the following conditions:

$$U_g(Y^*; X^*) \geq U_g\left(y_g, Y_{-g}^*; X(y_g, Y_{-g}^*)\right), \quad \forall g \in G,$$
$$U_u(Y^*; X^*) \geq U_u\left(Y^*; X_u, X_{-u}^*)\right), \quad \forall u \in U.$$

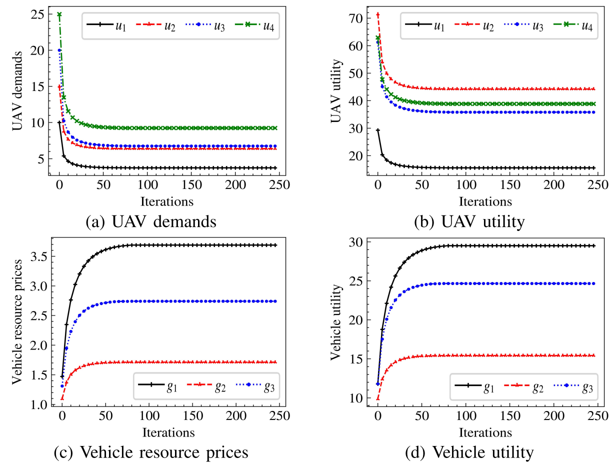
Theorem 2: The proposed game has a unique Stackelberg equilibrium.

Proof: In the first stage, the vehicles determine the price strategy Y. According to Theorem 1, a Nash equilibrium solution exists for the price optimization game among the vehicles. In the second stage, UAVs determine the optimal demand for computing resources using the proposed algorithm based on the price set by the vehicles in the first stage and their limited budgets. Therefore, a unique Stackelberg equilibrium solution exists for the proposed game. ∎

We find the Stackelberg equilibrium, where vehicles maximize their utility without knowing the prices for other vehicles, using the iterative algorithm in Algorithm 2. In Algorithm 2, the number of outer loop executions is ϑ/δ, where $\vartheta = |y_g(0) - y_g^*|$, and $y_g(0)$ represents the initial price. The number of inner loop executions is $|U||G| + |G|$. Therefore, the complexity of Algorithm 2 is $O(\vartheta(|U||G| + |G|)/\delta)$.

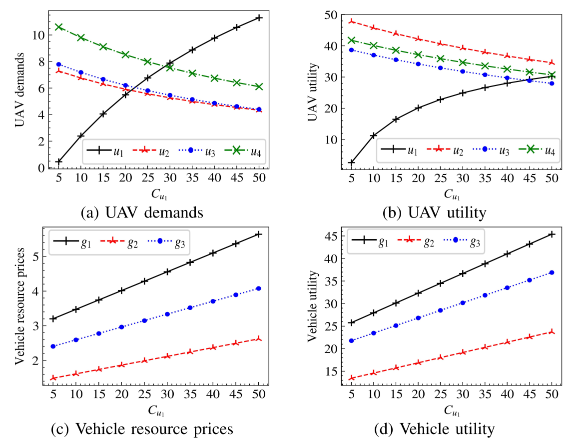
IV. Performance Evaluation

In the simulation, four UAVs, denoted by u_1, u_2, u_3, and u_4, are randomly placed within a 500 × 500 m area. Three vehicles, denoted by g_1, g_2, and g_3, are randomly placed within the communication range of the UAVs. The idle computing resources of the UAVs and vehicles are randomly selected from 1–3 and 5–10 GHz, respectively [11]. The altitude of each UAV was randomly set within a range of 100–150 m above ground. The data size s_u and required processing capacity ρ_u range from 50–100 Mbits and 100–150 cycles/bit, respectively [12]. The budget for each UAV is set as follows: $C_{u_1} = 10$, $C_{u_2} = 15$, $C_{u_3} = 20$, and $C_{u_4} = 25$(GHz × price). The channel bandwidth B and UAV transmission power $P_{u,g}$ are 10 MHz and 2 W, respectively. The Rician factor K is set to 10 [13]. We set the power gain at the reference distance σ and the noise power θ_0 to -50 dB and -110 dBm, respectively [7].



(a) UAV demands (b) UAV utility (c) Vehicle resource prices (d) Vehicle utility

Fig. 1. Convergence of the proposed algorithm.



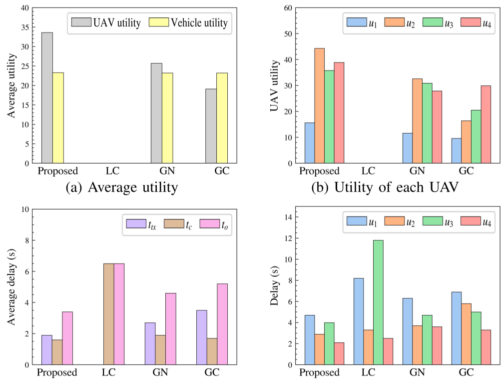
(a) UAV demands (b) UAV utility (c) Vehicle resource prices (d) Vehicle utility

Fig. 2. Performance analysis for budget of UAV u_1, C_{u_1}.

To evaluate the proposed scheme, we compared its performance with that of three benchmark schemes: (1) local computing (LC), wherein the task of each UAV is executed locally on the UAV itself; (2) greedy nearest (GN), wherein the task is offloaded from the nearest vehicle to the farthest [14]; and (3) greedy computing resources (GC), wherein the task is offloaded to the vehicles in order of the computing resources remained, from the most to the least [15], [16].

Fig. 1 illustrates the convergence performance of the proposed algorithm. The UAV demands and resource prices of vehicles converge after 90 iterations, as indicated in Figs. 1(a) and (c). Specifically, the UAV demands decrease, whereas the vehicle resource prices increase up to 50 iterations. As the iterations progress, the vehicles increase their resource prices to optimize their utility, leading to a decrease in the UAV demands. In addition, the convergence of the utilities of UAVs and vehicles, as depicted in Figs. 1(b) and (d), indicates that the proposed algorithm can reach a Nash equilibrium.

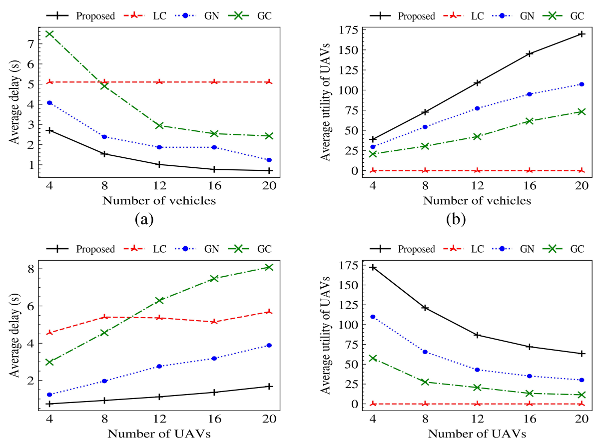
Fig. 2 shows the performance analysis under various budgets of UAV u_1. As the budget of u_1 increases, its demand also increases as it can buy more computing resources, as shown in Fig. 2(a). However, the higher demand of u_1 causes the vehicle



(a) Average utility (b) Utility of each UAV

(c) Average delay (d) Delay of each UAV

Fig. 3. Performance comparisons with benchmark schemes.



(a) (b)

(c) (d)

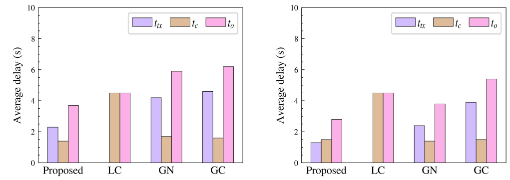
Fig. 4. Performance analysis for the numbers of UAVs and vehicles.

resource prices to increase, as in Fig. 2(c). Subsequently, the available computing resources of vehicles for UAVs other than u_1 start reducing, resulting in a decrease in their demands. Therefore, the utility of u_1 increases because it can acquire more computing resources, whereas that of the other UAVs decreases, as in Fig. 2(b).

Fig. 3 compares the performance of the proposed and benchmark schemes. The offloading delay t_u^o consisting of $t_{u,g}^{tx}$ and $t_{u,g}^{c}$ (as in Eq. (1)) is measured for each UAV to obtain the average delays t_o, t_{tx}, and t_c for all UAVs. The LC scheme has no utility or transmission delays for UAVs because all UAVs process tasks locally. Compared to GN and GC, the proposed scheme has 31 % and 76 % higher UAV utility, and 26 % and 35 % lower delays.

Fig. 4 illustrates the average delays and utility of UAVs as the numbers of UAVs and vehicles increase. As the number of vehicles increases, the average utility of the UAVs increases, and the average delay decreases in all schemes except LC because this allows UAVs to offload tasks to vehicles with lower transmission delays, whereas increased competition reduces the price of computing resources. When the number of UAVs increases, the utility of the UAVs decreases due to the increased demand for computing resources, raising the price of computing resources.

Fig. 5 compares the performance of random and uniform placements of UAVs [4]. Uniform placement leads to lower delays than random placement in all schemes except LC



(a) Random placement (b) Uniform placement

Fig. 5. Performance comparison with various UAV placements.

because it reduces the variation in distance between UAVs and vehicles, allowing more computing resources to be purchased.

V. Conclusion

This letter proposes a vehicle-aided task offloading strategy that aims to maximize the utility of UAVs and vehicles by modeling the task offloading process as a Stackelberg game between UAVs and vehicles. The simulation results reveal that the proposed offloading scheme outperforms other benchmarks.

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